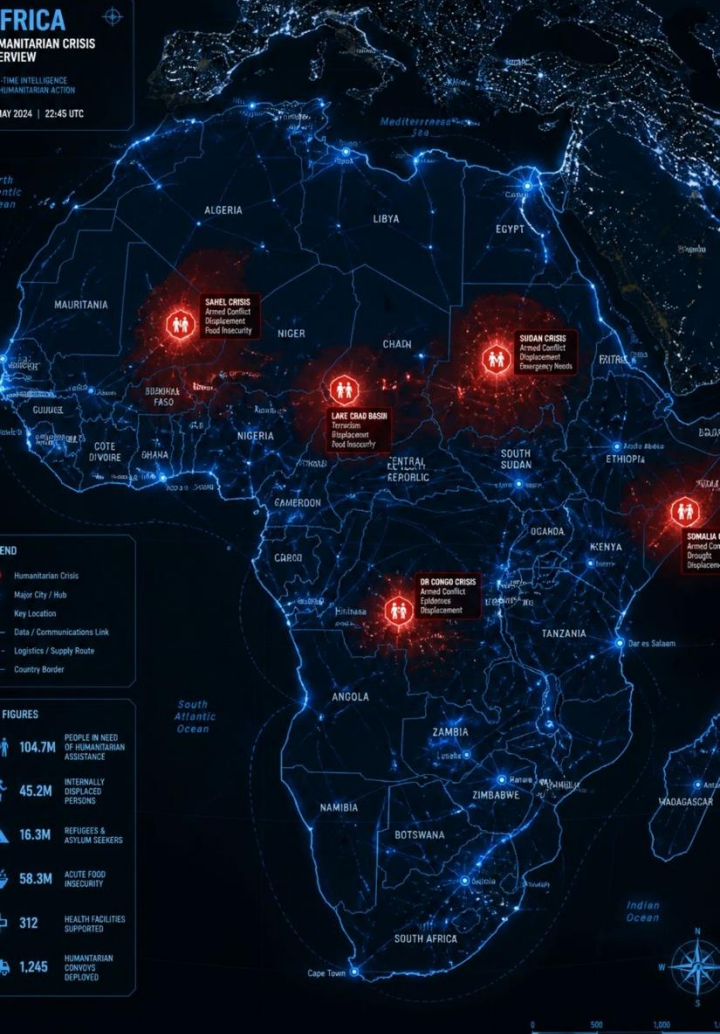




Humanitarian Data Ontology

A conceptual blueprint for building an **Admin 2 × Month** humanitarian data panel that preserves the provenance and meaning of each data source — linking conflict, climate, markets, and humanitarian reporting to food insecurity outcomes.

RESEARCH FRAMEWORK

ONTOLOGY DESIGN

ADMIN 2 × MONTH PANEL

This Week's Action Plan

Six prioritized tasks to complete the core humanitarian data panel prototype without waiting for the full ten-year, four-country pipeline.

1

Build One Prototype

Create an end-to-end Admin 2/month integrated table for one country-year using core sources: IPC, ACLED, CHIRPS, WFP prices, and ReliefWeb. Include GDELT only if the pipeline makes it easy.

2

Finalize the Ontology

Limit to four mechanisms: conflict, climate/agriculture, markets, and humanitarian reporting. Define no more than 3–5 features per mechanism.

3

Feature Specifications

For every feature, document: raw variable, source, spatial assignment rule, temporal aggregation rule, mathematical formulation, missing-value policy, and provenance/uncertainty flag.

4

Quantify Data Quality

Report: district-month coverage, missingness by source, proportion of exact vs. approximate geolocation, temporal availability, and duplicate or conflicting observations.

5

Select a Scientific Question

Preferably: *"Do different humanitarian sources provide complementary or redundant observations of the same crisis mechanisms?"*

6

Remove Non-Essential Sources

Place World Bank, IMF, and ICEWS in an optional extension list. They should not block completion of the core panel.

Ontology: Seven-Level Architecture

The main challenge is that each humanitarian dataset is built differently. They use different collection methods, spatial resolutions, and time scales. Because of that, we can't simply merge them together. The ontology provides a common framework that defines how each dataset is transformed into comparable features before everything is combined into a single integrated dataset.

01

Outcome of Interest

Food insecurity (IPC)

02

Humanitarian Mechanisms

Conflict, climate, markets, reporting

03

Observation Sources

ACLED, CHIRPS, WFP, ReliefWeb, VHI

04

Raw Variables

Extracted fields per source

05

Transformation Rules

Spatial and temporal processing

06

Engineered Features

Final indicators per mechanism

07

Integrated Data Panel

Admin 2 × Month unified dataset

Level 1: Outcome of Interest

Food Insecurity

The main outcome of interest is **food insecurity**, represented using the **Integrated Food Security Phase Classification (IPC)** — an internationally recognized framework classifying severity into five phases, from minimal (Phase 1) to catastrophe or famine (Phase 5).

Within the ontology, IPC is treated differently from all other datasets: it serves as the **reference outcome**, not an explanatory feature.

Conceptual Distinction

By placing food insecurity at the highest level, the ontology draws a clear distinction between the **outcome variable** and the **explanatory variables** derived from conflict, climate, markets, and humanitarian reporting datasets.

The remaining mechanisms may be *associated* with food insecurity but are not direct measurements of IPC classification

Level 2: Four Humanitarian Mechanisms

Each mechanism captures a distinct dimension of the humanitarian environment, avoiding the assumption that all datasets measure the same phenomenon.



Conflict & Insecurity

Armed conflict, political violence, and insecurity disrupting agricultural production, displacing communities, interrupting supply chains, and restricting humanitarian access.



Climate & Agricultural Stress

Environmental conditions — rainfall variability, droughts, and climate events — reducing crop yields, affecting livestock, and increasing food insecurity risk.



Market & Food Price Pressure

Rising food prices, market disruptions, and reduced market access limiting household purchasing power and increasing vulnerability to food insecurity.



Humanitarian Reporting & Response

Level of humanitarian activity and reporting associated with a crisis, providing context on severity and location — though reporting intensity is not a direct measure of need.

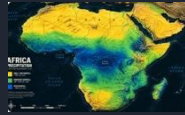
Level 3: Observation Sources

Mechanisms cannot be observed directly — they are represented through complementary datasets, each with its own collection methods, spatial resolution, and temporal frequency.



ACLED

Conflict & Insecurity — Records conflict events, political violence, protests, and fatalities. GDELT is an optional future extension for media-reported global news coverage.



CHIRPS

Climate & Agricultural Stress — Satellite-derived rainfall estimates for assessing rainfall variability and potential agricultural stress.



WFP Food Prices

Market & Food Price Pressure — Market-level prices for staple commodities, monitoring changes in food affordability and market conditions.



ReliefWeb

Humanitarian Reporting — Aggregates reports from governments, NGOs, and UN agencies on ongoing crises and humanitarian activities.

Vegetation Health Index (VHI)

The Vegetation Health Index (VHI) assesses vegetation health using satellite measurements of surface temperature and greenness.

It combines the Vegetation Condition Index (VCI), reflecting vegetation greenness, and the Temperature Condition Index (TCI), reflecting temperature. Low VHI values typically indicate vegetation stress, often linked to drought conditions.

- ❶ VHI serves as an early indicator for potential agricultural stress and food insecurity, offering vital insights in climate-vulnerable regions.



Level 4: Transformation Rules

All raw data is processed to a common **Admin 2 × Month** spatial-temporal framework.

Conflict (ACLED)

Conflict events are assigned to Admin2 areas using their coordinates. Events are then grouped by month, allowing us to calculate the number of each event type, total fatalities, and conflict density for every district.

Climate (CHIRPS & VHI)

Monthly rainfall and vegetation data are overlaid with Admin2 boundaries. We then calculate summary statistics, such as average rainfall and average vegetation health, for each district. Since both datasets are already monthly, no additional time aggregation is needed.

Markets (WFP VAM)

Market observations are assigned to Admin2 areas using their coordinates. Prices are averaged by commodity and month, while keeping each commodity separate to reflect local food consumption patterns across countries.

Reporting (ReliefWeb)

ReliefWeb reports are grouped by country and month. Since most reports do not provide reliable Admin2 locations, reporting indicators are calculated at the country level rather than for individual districts.

Level 5: Raw Variables

Each source contributes specific raw variables extracted for feature engineering.

Mechanism	Source	Key Raw Variables
Conflict & Insecurity	ACLED	event_date, event_type (6 categories), latitude, longitude, fatalities, country, admin1
Climate & Agricultural Stress	CHIRPS	Monthly rainfall (mm)
Climate & Agricultural Stress	VHI	Monthly Vegetation Health Index
Markets & Food Price Pressure	WFP Food Prices	Market, commodity, price, currency, unit, date, latitude, longitude
Humanitarian Reporting	ReliefWeb	Date, primary country, themes (Food & Nutrition, Health, WASH, etc.), format

📄 World Bank, IMF, and ICEWS are placed in an optional extension list – they do not block completion of the core panel.

Feature specification: Conflict & Insecurity

Feature	Description	Mathematical formulation	Spatial Assignment rule	Temporal aggregation rule	Missing Value Policy
conflict_event_count	Total number of conflict events occurring within an Admin 2 unit during the month.	Count of all conflict events within an Admin2 district during the month.	Assign each event to the corresponding Admin2 using ACLED coordinates.	Monthly count of all events	Missing values are set to 0
battles_count	Number of events classified as Battles.	Count of events where Event Type = "Battles".	Events grouped by Admin 2 administrative unit	Monthly sum	Missing values are set to 0
explosions_remote_violence_count	Number of events classified as Explosions/Remote Violence.	Count of events where Event Type = "Explosions/Remote Violence".	Events grouped by Admin 2 administrative unit 2	Monthly sum	Missing values are set to 0
violence_against_civilians_count	Number of events classified as Violence against Civilians.	Count of events where Event Type = "Violence Against Civilians".	Events grouped by Admin 2 administrative unit	Monthly sum	Missing values are set to 0
riots_count	Number of events classified as Riots.	Count of events where Event Type = "Riots".	Events grouped by Admin 2 administrative unit	Monthly sum	Missing values are set to 0
protests_count	Number of events classified as Protests.	Count of events where Event Type = "Protests".	Events grouped by Admin 2 administrative unit	Monthly sum	Missing values are set to 0
strategic_developments_count	Number of events classified as Strategic Developments.	Count of events where Event Type = "Strategic Developments".	Events grouped by Admin 2 administrative unit	Monthly sum	Missing values are set to 0
fatalities	Total reported fatalities associated with all conflict events during the month.	Sum of fatalities reported across all conflict events within the Admin2 district during the month	By Admin 2 level	Monthly sum	Missing values are set to 0
conflict_density	Number of conflict events per square kilometre of the administrative unit.	Conflict Event Count / Area of Admin2 (km2)	Area calculated from GADM Admin 2 polygons	Monthly	Missing values are set to 0
conflict_trend_pct	Percentage change in the total number of conflict events compared with the previous month.	(Current Month Event count - Previous Month Event count / previous Month Event count) * 100	By Admin 2 level	Monthly percentage change	Missing values are set to 0

Feature specification: Climate & Agricultural Stress

Feature	Description	Mathematical formulation	Spatial Assignment rule	Temporal aggregation rule	Missing Value Policy
vegetation_health_index_mean	Average Vegetation Health Index within an Admin 2 unit in the month.	Mean VHI across all raster cells within Admin 2 boundary	Mean VHI raster values for each admin 2 polygon	Monthly average.	Retain NaN if raster data are unavailable.
rainfall_max_mm	Highest monthly rainfall recorded within an Admin2 district (mm).	Maximum rainfall value across all CHIRPS raster cells within the Admin2 boundary.	Maximum CHIRPS raster value within each Admin2 administrative unit.	Monthly maximum.	Retain NaN if raster data are unavailable.
rainfall_min_mm	Lowest monthly rainfall recorded within an Admin2 district (mm).	Minimum rainfall value across all CHIRPS raster cells within the Admin2 boundary.	Minimum CHIRPS raster value within each Admin2 administrative unit.	Monthly minimum.	Retain NaN if raster data are unavailable.
rainfall_mean_mm	Average monthly rainfall within an Admin2 district (mm).	Mean rainfall across all CHIRPS raster cells within the Admin2 boundary.	Mean CHIRPS raster values calculated for each Admin2 administrative unit using zonal statistics.	Monthly average.	Retain NaN if raster data are unavailable.

Feature specification: Markets & Food Price Pressure

Feature	Description	Mathematical formulation	Spatial Assignment rule	Temporal aggregation rule	Missing Value Policy
commodity_price	Average monthly price of a selected staple commodity within an Admin2 administrative unit.	Average price of all market observations for the same commodity within an Admin2 administrative unit during a month.	Each market observation is assigned to an Admin2 administrative unit using its latitude and longitude coordinates	Market observations are grouped by Admin2 administrative unit, commodity, and calendar month.	Market observations with missing prices or invalid coordinates are excluded.
commodity_price_change_pct	Percentage change in the commodity price compared with the previous month.	$((\text{Current Month Price} - \text{Previous Month Price}) \div \text{Previous Month Price}) \times 100$	Calculated using monthly average commodity prices for the same Admin2 administrative unit.	Calculated from consecutive monthly commodity prices after monthly aggregation.	Recorded as NA when either the current or previous month's price is unavailable.

Feature specification: Humanitarian reporting

Feature	Description	Mathematical formulation	Spatial Assignment rule	Temporal aggregation rule	Missing Value Policy
humanitarian_report_count	Number of ReliefWeb reports related to food insecurity within an Admin2 administrative unit during a month.	Count of ReliefWeb reports assigned to the same Admin2 administrative unit during a calendar month.	Reports are geocoded using the location information provided in ReliefWeb. Each report is assigned to an Admin2 administrative unit based on its reported location.	Reports are grouped by Admin2 administrative unit and calendar month. Each qualifying report contributes one count.	If no qualifying reports exist for an Admin2 administrative unit during a month, the feature is assigned a value of 0.
report_theme_count	Number of reports classified under each food insecurity theme (e.g., conflict, displacement, nutrition, food assistance).	Count of reports belonging to each thematic category within an Admin2 administrative unit during a month.	Reports are assigned to an Admin2 administrative unit using their reported location. Theme labels are retained from the ReliefWeb metadata or ontology classification.	Reports are grouped by Admin2 administrative unit, theme, and calendar month.	If no reports exist for a given theme during a month, the feature is assigned a value of 0.

Level 7: Integrated Humanitarian Data Panel

The final ontology level combines all engineered features into a single **Admin 2 × Month** dataset — each record represents one administrative area during one month, with features from all four mechanisms unified while preserving source provenance.

Core Scientific Question

"Do different humanitarian sources provide complementary or redundant observations of the same crisis mechanisms?"

Humanitarian Intelligence Dashboard

INTEGRATED DATA. ACTIONABLE INSIGHTS. BETTER OUTCOMES.

20 MAY 2024
DATA AS OF 18 MAY 2024 23:00
NEXT UPDATE IN 23H



Next steps: Building prototype

Common Schema

record_id
month
year
country
admin 1
admin 2
latitude
longitude

Conflict & Insecurity

conflict_event_count
event_count
conflict_trend_pct
fatalities

Market & Food Price Pressure

commodity_price
commodity_price_change_pct

Climate / Agriculture

rainfall_mean_mm
rainfall_min_mm
rainfall_max_mm
vegetation_health_index_mean

Humanitarian Reporting

humanitarian_report_count
report_theme_count